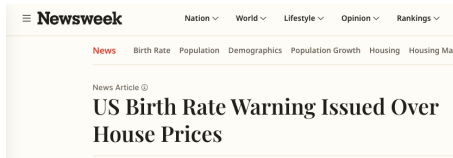


Fertility and Housing Tenure Choice

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September 14, 2026

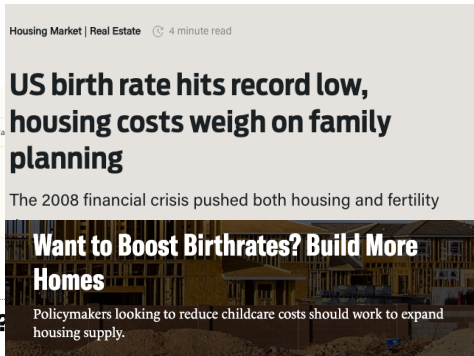
Housing and Fertility



POLICY

Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.



Renting a home 'leads to lower birth rates'

Housing, Fertility, and Tenure Choice

Children increase how much physical space a household needs. Access to that space depends on tenure and wealth:

- Rental housing can limit the size of the home a family can occupy.
- Ownership expands the housing menu but requires liquid wealth for a down payment.
- Older owners may retain family-sized homes after their children leave.
- Young families and older households compete for housing: its allocation across generations matters for fertility.

How much does access to family-sized housing affect fertility?

Which housing constraints matter, and for whom?

This Paper

Need a framework where fertility, tenure, housing size, saving, and population are jointly determined.

Model: overlapping generations with fertility, tenure, and saving

- Income heterogeneity, earnings risk, and bequests.
- Rental size limits, down payments, and housing transaction costs.
- Childbirth raises housing needs and changes future housing demand.

Empirics: housing around births, birth timing, and wealth by age

Quantification: calibrate the model and study the demographic transition

- Discipline the initial economy and the decline in fertility preferences.
- Compare housing policies from the same inherited transition state.

Quantitative Model

Environment

- Overlapping generations of households in one competitive housing market.
 - Rental space is size-limited; ownership offers larger homes but requires a down payment.
- Households age from $a = 0$ to J , survive with probability s_a , and retire at a_R .
 - Birth attempts are possible during a fertile window $[a_F^0, a_F^1]$.
 - Children depend on their parents until they mature.
- Households differ in liquid wealth b , inherited owned housing h , and persistent earnings z .
 - Lifetime parity n and dependent children m describe their family composition.
- Households choose birth attempts, tenure, housing size, consumption, and saving.

Preferences

Households value consumption c , housing services s , and children at home m :

$$u_t(c, s; m) = e(m)^{\sigma-1} \frac{\left[c^{\alpha_0} (s - \bar{h}(m))^{1-\alpha_0} \right]^{1-\sigma}}{1-\sigma} + \psi_t m.$$

Family size raises household needs through the equivalence scale $e(m)$ and the space requirement $\bar{h}(m)$.

- α_0 is the consumption share; σ governs curvature.
- ψ_t is the preference for children at date t .
- The equivalence scale $e(m)$ (Scholz et al., 2006) is increasing and concave.
- Its shape is fixed externally and normalized to a childless couple.

Family Space

The first child creates a discrete increase in required housing services:

$$\bar{h}(m) = h_P \mathbf{1}\{m > 0\}.$$

h_P is the housing requirement while dependent children are present.

Housing services depend on tenure:

$$s = \begin{cases} h^R, & \text{renter,} \\ \chi h', & \text{owner,} \end{cases} \quad \chi > 0.$$

The multiplier χ converts owned rooms into housing services; $\chi > 1$ gives an ownership premium.

Child needs disappear as children mature; lifetime births remain in the state.

Earnings and Bequests

Gross earnings $y_t^g(a, z)$ combine an age profile and persistent income risk (De Nardi, 2004). Earnings state z follows a Markov process; households receive pensions after working life.

At death, households value estate wealth w^e :

$$\mathcal{B}(w^e) = \theta_0 \frac{(\theta_1 + \max\{w^e, 0\})^{1-\sigma}}{1-\sigma}.$$

θ_0 governs the bequest motive; θ_1 governs its sensitivity to wealth.

- Estate wealth includes liquid assets and housing wealth.
- Bequests affect saving and old-age housing retention.
- Entrant wealth is externally distributed; the model is not fully dynastic.

Pensions

A fixed payroll tax τ^{SS} finances pensions $\varpi_t d(z)$, with retirement at age a_R :

$$\tau^{SS} \int_{a < a_R} y_t^g(a, z) dG_t = \varpi_t \int_{a \geq a_R} d(z) dG_t.$$

The pension ϖ_t balances the budget using the actual household distribution G_t . The benefit multiplier $d(z)$ equals one for common benefits.

Disposable earnings and pension income are:

$$y_t^d(a, z) = \begin{cases} (1 - \tau^{SS}) y_t^g(a, z), & a < a_R, \\ \varpi_t d(z), & a \geq a_R. \end{cases}$$

Households anticipate the pension path. Earnings and pensions are measured over the model period.

Property-tax rebates T_t have a separate budget.

Housing and Tenure

Rental housing is continuous up to a size limit; owned housing comes in discrete sizes:

$$h^R \in [0, \bar{h}^R], \quad h' \in \{0\} \cup \mathcal{H}^O, \quad h' h^R = 0.$$

A household either rents or owns, and the largest owned homes exceed the rental limit.

- Buying requires a down payment of $(1 - \phi)P_t h'$.
- P_t is the purchase price per room; ϕ is the financed share.
- Selling incurs a proportional transaction cost ψ^S .
- Borrowing is secured by housing; the debt limit tightens with age.

Budget Constraints

Changing inherited owner size h to h' changes liquid resources by

$$\mathcal{T}_t(h, h') = \mathbf{1}\{h' \neq h\} [(1 - \psi^s)P_t h - P_t h'].$$

Housing transactions change liquid wealth before financial returns accrue.

For renters and owners, respectively,

$$c + r_t h^R + b' = R_b [b + \mathcal{T}_t(h, 0)] + y_t^d(a, z) + T_t,$$

$$c + (\delta_H + \tau_t^P)P_t h' + b' = R_b [b + \mathcal{T}_t(h, h')] + y_t^d(a, z) + T_t.$$

R_b is the gross financial return per period, r_t rent per room, δ_H depreciation, τ_t^P the property-tax rate, and T_t a transfer per household head.

Households carry net liquid wealth b into the period and choose saving b' .

Sequential Fertility and Child Aging

- During fertile ages, a household below terminal parity chooses whether to attempt the next birth.
- An attempt succeeds with age-specific probability π_a .
- A successful birth:

$$n' = n + 1, \quad m' = m + 1,$$

and the first successful birth pays the one-time utility cost F_1 .

- Fertility taste shocks have scale κ_1 on the first-birth margin and κ_C on continuation margins.
- Each dependent child matures independently; the state tracks their count m , while lifetime parity $n \in \{0, 1, 2, 3+\}$ remains a stock.

A birth changes the household's space needs immediately, but its demographic effects propagate across cohorts.

Housing Supply and Prices

Physical housing supply responds to the purchase price:

$$H_t^s = H_0 \left(\frac{P_t}{\bar{P}} \right)^\eta.$$

H_0 is supply at reference price $\bar{P}$; η is the supply elasticity.

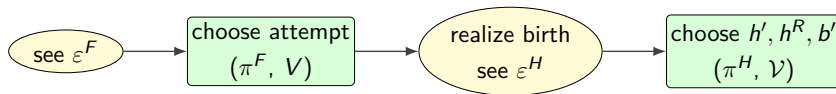
Rent and purchase prices satisfy the asset-pricing condition:

$$r_t + P_{t+1} = (R_b + \delta_H + \tau_t^p)P_t.$$

The current rent depends on both the carrying cost and the anticipated resale price.

Housing is supplied separately at each date; there is no construction stock in this specification.

The Household's Problem: Sequential Choices



At date t and age a , write the household's inherited state as

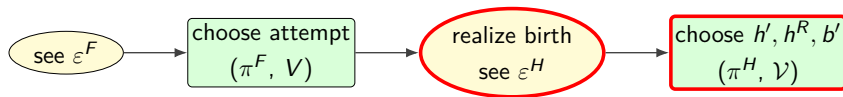
$$x = (b, h, z, n, m).$$

b is liquid wealth, h owned housing ($h = 0$ for renters), z the earnings state, n lifetime parity, and m dependent children.

$V_{t,a}(x)$ is value before fertility choice; $\mathcal{V}_{t,a}(x)$ is value after the birth outcome, before housing choice.

A successful birth changes the state to $x^+ = (b, h, z, n + 1, m + 1)$. Households then choose housing and saving for the realized family size.

The Household's Problem: Housing and Saving

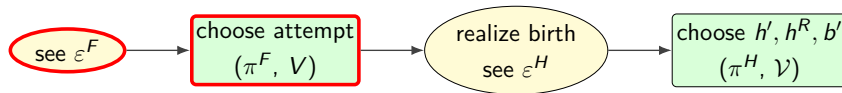


After the birth outcome, housing and saving solve

$$\begin{aligned} \mathcal{V}_{t,a}(b, h, z, n, m) = \mathbb{E}_{\varepsilon_t^H} \max_{h', h^R, b'} \bigg\{ & u_t(c, s; m) + \varepsilon_t^H(h') \\ & + \beta s_a \mathbb{E}[V_{t+1,a+1}(b', h', z', n, m') \mid z, m] \\ & + \beta(1 - s_a) \mathbb{E}_t[\mathcal{B}(w_{t+1}^e)] \bigg\}. \end{aligned}$$

- Choices satisfy budgets, the rental size cap, down-payment requirements, and age-dependent debt limits.
- Expectations integrate next period's earnings z' and dependent-child count m' .
- β discounts the future, s_a is survival, and κ_H is the scale of housing tastes.

The Household's Problem: Fertility Choice



Given conception success π_a , waiting and attempting have values

$$W_{t,a}^0(x) = \mathcal{V}_{t,a}(x), \quad W_{t,a}^A(x) = \pi_a [\mathcal{V}_{t,a}(x^+) - F_1 \mathbf{1}\{n = 0\}] + (1 - \pi_a) \mathcal{V}_{t,a}(x).$$

The value before fertility tastes and the attempt probability are

$$V_{t,a}(x) = \mathbb{E}_{\varepsilon^F} \max\{W_{t,a}^0(x) + \varepsilon_0^F, W_{t,a}^A(x) + \varepsilon_A^F\},$$

$$\pi_{t,a}^F(x) = \frac{\exp\{W_{t,a}^A(x)/\kappa(n)\}}{\exp\{W_{t,a}^A(x)/\kappa(n)\} + \exp\{W_{t,a}^0(x)/\kappa(n)\}}, \quad \kappa(n) = \begin{cases} \kappa_1, & n = 0, \\ \kappa_C, & n \geq 1. \end{cases}$$

Expected births are $\pi_a \pi_{t,a}^F(x)$. Attempts are unavailable outside fertile ages or at terminal parity.

Population and Households

Births change future housing demand by changing the number and age of households.

- N_t : resident persons, tracked by age and sex.
- H_t : household heads, obtained from persons using age–sex headship rates.
- G_t : the distribution of household wealth, earnings, tenure, and children.

Births enter the population immediately. Survival, migration, and household formation determine when they add to housing demand.

Equilibrium

An equilibrium is a sequence of policy functions, value functions, household distributions, populations, prices, transfers, and pensions such that:

1. Given prices, fiscal paths, and expectations, policies solve the household's problem and values give the associated lifetime utilities.
2. Annual population accounting and fixed headship rates satisfy:

$$N_{y+1} = A_{y+1}N_y + f_{y+1}B_{y+1} + M_{y+1}, \quad H_y^{g,a} = \rho^{g,a}N_y^{g,a}.$$

Fertility policies determine B ; A ages survivors, fB counts surviving newborns, and M is net migration.

3. From the initial state, distributions follow policies, idiosyncratic shocks, and demographic entry and exit, with $\int dG_t = H_t$, the head count over represented adult ages.
4. Housing clears, $\int q_t dG_t = H_t^s$, with q_t measured in physical rooms. Asset pricing and both fiscal budgets hold.

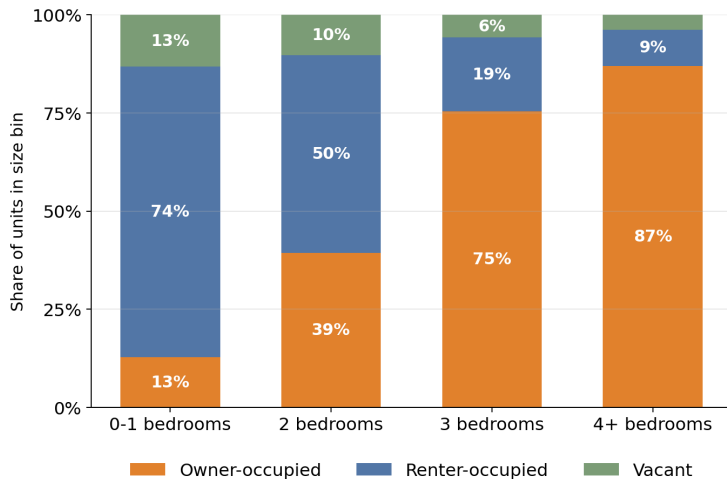
Empirics

Data

- **AHS and ACS:** housing sizes, tenure, household composition, and age profiles.
- **PSID:** housing around childbirth, earnings, and wealth by age.
- **CPS and natality records:** completed fertility, childlessness, and birth timing.
- **Census population estimates:** age–sex population and demographic flows.

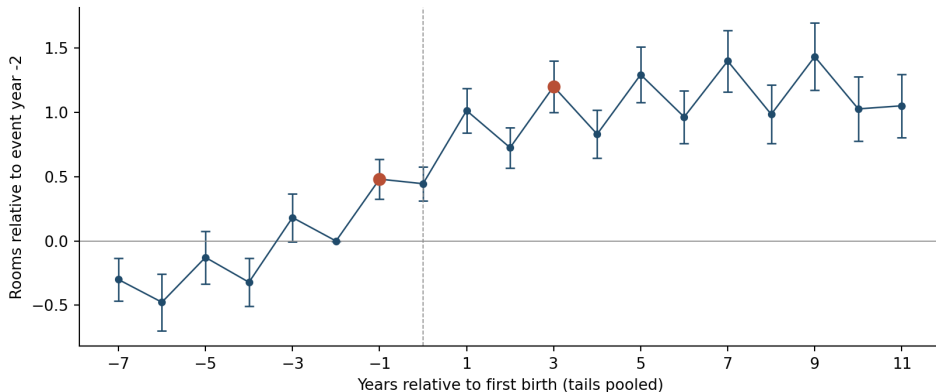
Cross-sectional housing patterns discipline access to space; longitudinal data describe housing adjustment around births.

Big Units Are Owned



Source: American Housing Survey, 2023.

Housing Size Around First Birth



Provisional event profile: empirical timing and reference-period support are under review. The retained calibration contrast is -1 to $+3$; it is not a causal effect.

Source: PSID; Sun-Abraham event study, household-aligned sample; pointwise 95% intervals.

Quantification

1. Calibrating the Initial Economy

Goal: fit pre-2007 household moments. External inputs and all 12 targets and weights stay fixed.

1. **Try a parameter vector θ :** nine coefficients governing saving, fertility, housing, and bequests.
2. **Compute initial age–earnings shares** from survival, entry, and the earnings process. Set the pension to payroll revenue divided by retiree exposure; verify it against the solved distribution.
3. **Try a fertility preference ψ_0 and solve the housing equilibrium.** Guess the house price P , derive rent, solve household choices backward, and find the stationary distribution forward. Update P and repeat until housing demand equals supply.
4. **Update ψ_0 with a bracketed scalar search.** Repeat step 3 for each trial until model completed fertility equals 2.1 (within 0.0005).
5. **Score this parameter vector:** compute all 12 model moments and the weighted sum of squared model–target gaps.
6. **Update θ .** Perturb each coefficient to estimate how the moments respond; use these slopes for bounded, ridge-regularized joint steps. Redo steps 2–5 for each proposal, keep the lowest loss, and repeat within the search budget. Re-solve the selected winner twice.

Nested loops: parameter search $\longrightarrow$ fertility normalization $\longrightarrow$ housing-market clearing.

2. Fitting the Historical Transition

Hold the nine structural parameters fixed. Form 2007 age masses from data; within each age, keep the calibrated distribution of wealth, housing, and family states.

1. **Try ψ_{2007} , believed permanent.** Solve the terminal stationary economy at this preference level.
2. **Guess future price and pension paths.** Solve household choices backward from the terminal value; simulate household and person distributions and birth cohorts forward from the inherited 2007 state.
3. **Update the price and pension paths and repeat step 2** until housing clears and the payroll budget balances at every date. Verify the population accounting, recompute the path, and check a longer horizon.
4. **Change ψ_{2007} and repeat steps 1–3** until first-window fertility matches the 2008–2011 target of 1.974875 (error at most 0.005). Every trial starts from the same 2007 state.
5. **Advance only after acceptance.** Keep the first four-year outcome and birth queues. Repeat at 2011, 2015, and 2019 to target the following four years. Each new preference level is unanticipated.

After 2023, hold the final preference level fixed.

The household-rate analogue of female TFR is approximate; the complete history is not yet validated.

Empirical Discipline

Moments inform parameters through their joint effects on household choices.

Margin	Empirical variation	Parameters
Fertility	Childlessness and one-child families; mean first-birth age and the share at age 30+	F_1, κ_1, κ_C
Housing	Rooms and ownership; housing around first birth and differences across family sizes	h_P, χ, H_0
Saving	Wealth relative to earnings, bequest flows, and old-age wealth dispersion	$\beta, \theta_0, \theta_1$

Fertility stocks and birth timing discipline different margins. Housing around birth informs demand for space.

Moment Construction

Compare model moments with the same age windows, conditioning events, and time units used in the data.

- **Fertility:** CPS 2004/2006 parity stocks at ages 40–44; natality records 2003–2006 for first-birth timing. Interpolate birth timing within model age cells.
- **Housing:** ACS 2005/2006, pooled across 42 metros; match room top-codes, age ranges, and resident-child groups. The model uses dependent counts as the child-group proxy.
- **Wealth:** PSID 2003/2005; match weighted wealth/earnings and old-age wealth/income dispersion. Bequests/wealth is an external flow restriction.
- **Housing around birth:** compare a longitudinal PSID room response with the corresponding model event contrast; the empirical timing and reference design remain under review.

Fertility Targets

Provisional initial calibration

Moment	Target	Model
Completed fertility (normalization)	2.10	2.10
Childless, women 40–44 (%)	19.8	20.7
Exactly one child among mothers, 40–44 (%)	21.4	20.7
Mean age at first birth (years)	25.98	26.10
First births at age 30+ (%)	24.9	23.2

Source: CPS fertility supplements, 2004/2006; NCHS natality, 2003–2006. Completed fertility 2.1 is an author-selected model normalization.

Housing and Wealth Targets

Provisional initial calibration

Moment	Target	Model
Mean occupied rooms (capped at 9)	5.56	6.29
Ownership, heads 30–55 (%)	64.8	52.4
First-birth room response, –1 to +3 (†)	0.720	1.011
Rooms: 3+ versus 1–2 resident children	0.347	0.200
Recent-parent ownership gap (pp)	16.3	15.0
Wealth / annual gross earnings	6.15	4.85
Annual bequests / wealth (%)	0.88	0.86
Wealth/income p90 / median, ages 76–84	3.52	4.44

† Retained computational target; empirical measurement under review.

Source: ACS 2005/2006; PSID 2003/2005 wealth and pooled first-birth event study; external bequest-flow restriction.

Calibrated Parameters

Provisional initial calibration

Parameter	Economic interpretation	Value
β_{annual}	Annual discount factor	0.990
κ_1	First-birth taste dispersion	0.279
κ_C	Later-birth taste dispersion	0.347
χ	Owner housing-service multiplier	1.048
H_0	Housing supply at reference price	8.191
θ_0	Bequest-motive scale	0.087
θ_1	Bequest-utility wealth shift	0.057
F_1	First-birth utility cost	0.127
h_P	Parenthood space requirement	2.300

β_{annual} and h_P reach their upper search bounds.

► Parameter restrictions

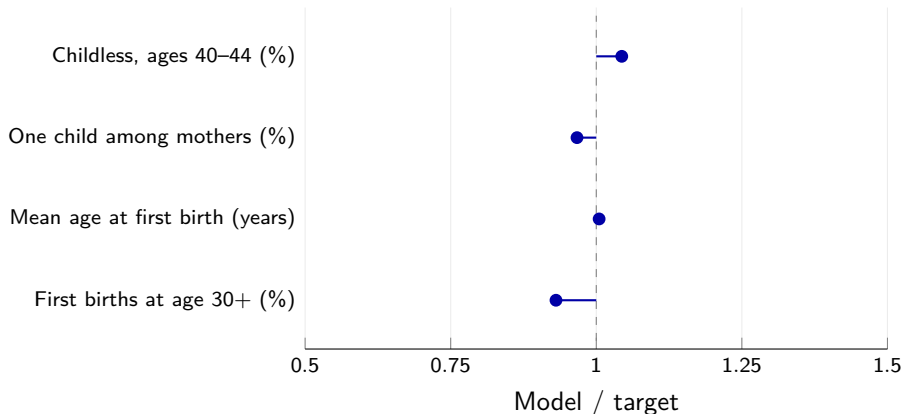
External Inputs and Normalizations

Object	Economic interpretation	Value
σ	Utility curvature (fixed)	2
α_0	Consumption share (fixed)	0.733
η	Housing-supply elasticity (fixed)	0.63
κ_H	Housing-product taste dispersion (fixed)	0.005
τ^{SS}	Payroll tax rate (fixed)	0.179
Child-space slope	No additional linear requirement per child	0
ψ_0	Preference level matching completed fertility 2.1	0.1596
ϖ_0	Initial pension implied by budget balance	2.0464

A model period is four years, $\beta = \beta_{\text{annual}}^4$. The equivalence scale is $e(m) = ((2 + 0.7m)/2)^{0.7}$; pensions are in model-period income units.

Fertility Fit

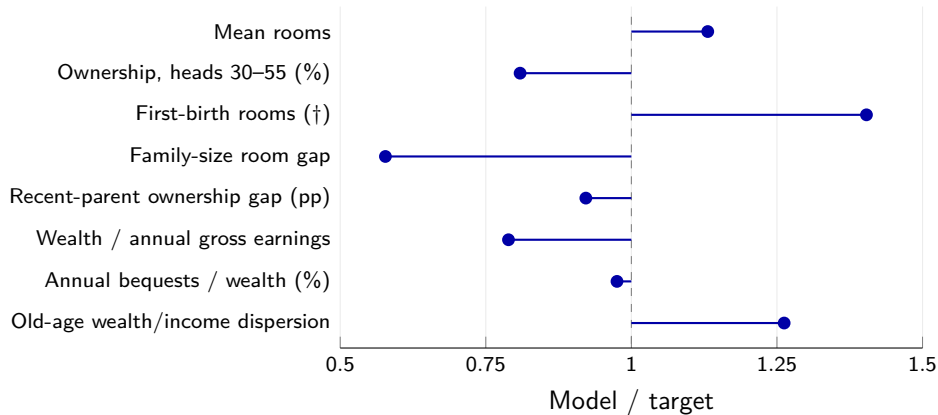
Provisional initial calibration



The dashed line denotes the target. First births occur slightly later in the model, but too few occur at age 30 or above.

Housing and Wealth Fit

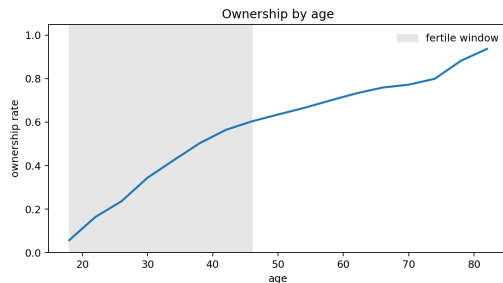
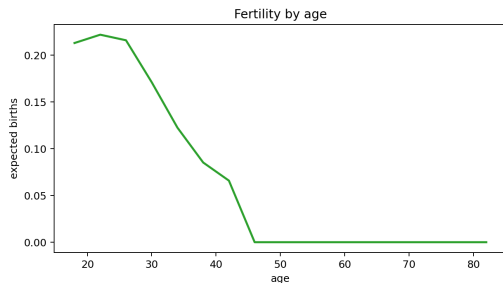
Provisional initial calibration



The model has too many rooms and too little ownership and wealth. † The first-birth room target awaits empirical reconciliation.

The Initial Equilibrium over the Lifecycle

Provisional initial calibration



Births are concentrated early in life, while ownership rises with age.

These are model equilibrium profiles; the preceding tables and plots compare the targeted moments with data.

Historical Fertility Discipline

Choose each preference level to match fertility over the following four calendar years.

Decision date	Birth years	Mean published TFR
2007	2008–2011	1.9749
2011	2012–2015	1.8610
2015	2016–2019	1.7554
2019	2020–2023	1.6458

The model sums age-specific births per adult household: an approximate analogue of female period TFR. The exposure mapping remains to be reconciled.

This is the estimation design. A complete historical fit requires converged conditional equilibria and stability to a longer continuation horizon.

Source: Published annual U.S. total fertility rates; equal-weight four-year averages.

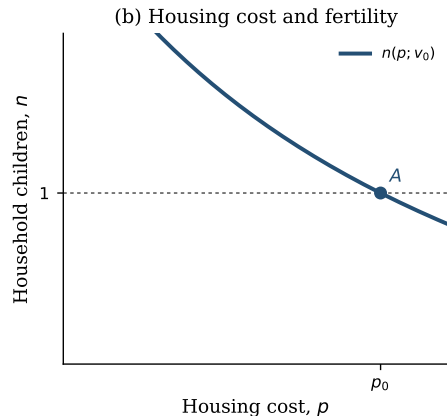
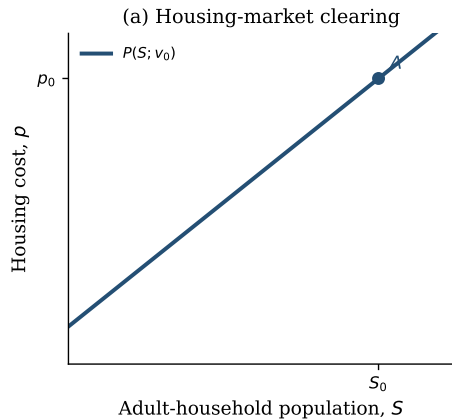
Housing and Fertility in Equilibrium

The next three diagrams illustrate the interaction between housing costs and population.

- **Left panel:** $P(S; v)$ is the housing cost that clears the market at adult-household population S , holding age composition fixed.
- **Right panel:** $n(p; v)$ is household fertility at housing cost p and fertility preference v .
- The illustration uses a closed population, with replacement equal to $n = 1$.

These are schematic schedules. The quantitative model tracks the full age distribution and distinguishes rents from asset prices.

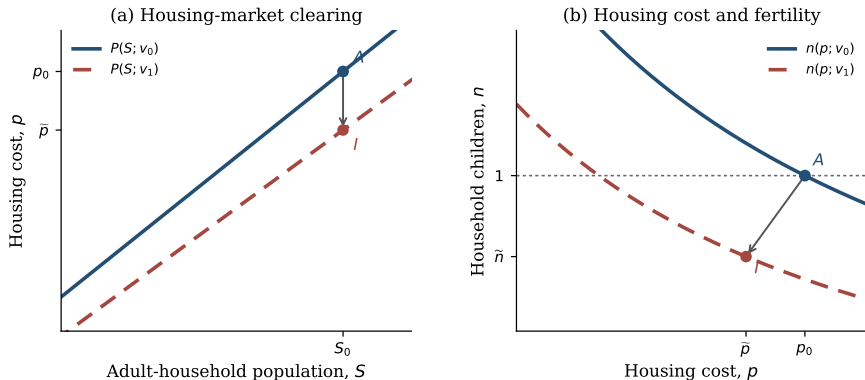
Initial Steady State



At A , housing clears at (S_0, p_0) and fertility is at replacement.

The housing cost and population are constant in this schematic initial economy.

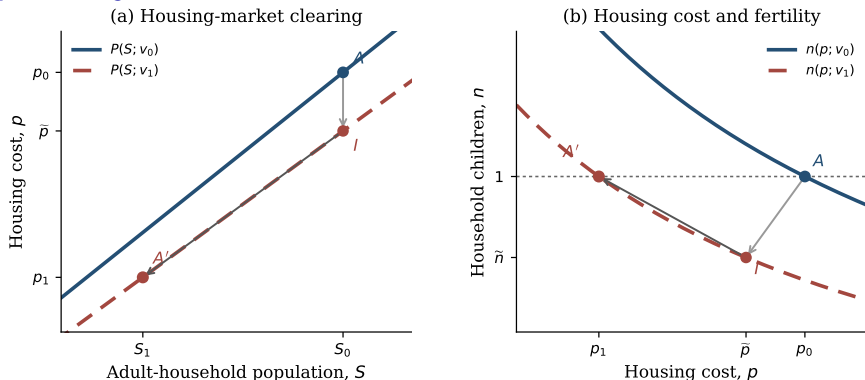
Impact



A fall from v_0 to v_1 shifts both schedules. With S_0 inherited, the schematic impact equilibrium moves from A to I : housing cost and fertility fall.

The diagram shows a one-time shift; the quantitative preference decline is spread over time.

Demographic Adjustment



Smaller entering cohorts can reduce housing demand. Lower housing costs partly offset the preference decline; A' marks a possible stationary outcome.

The arrow links the illustrated equilibria. Changing age composition can produce overshooting; it is not a computed transition or a convergence result.

Policy Along the Transition

At date t_p , introduce a housing reform after the fertility-preference decline has begun.

Both paths start from the same state immediately before that announcement:

$$(G_{t_p}, N_{t_p}, H_{t_p})^{\text{reform}} = (G_{t_p}, N_{t_p}, H_{t_p})^{\text{baseline}}.$$

The inherited liquid assets, homes, and age distributions are therefore identical.

- Keep preferences, survival, migration, and the entry rule common.
- Preserve the housing-supply schedule except when it is the policy instrument.
- Solve a new anticipated price and allocation path after the announcement.

The Rebated Property Tax

Consider a permanent property-tax increase with contemporaneous revenue returned equally to household heads:

$$\tau_t \int dG_t = \tau_t^P P_t Q_t^H, \quad Q_t^H = \int q_t(x) dG_t(x).$$

Here $q_t(x)$ is occupied physical housing, including rented rooms; τ_t^P is the tax rate per model period.

- **Capitalization:** changes in carrying costs affect asset prices.
- **Access:** prices and transfers change down-payment affordability.
- **Household choices:** tenure, housing size, saving, and fertility adjust.
- **Demography:** different births change future cohorts and housing demand.

Compare paths under the same rebate rule, with a common inherited state and supply schedule.

Conclusion

- Children raise demand for rooms; rental size and down payments govern access to that space.
- Tenure, saving, and fertility interact over the household lifecycle.
- Births change population and housing demand with long delays.
- Housing policy changes choices along an ongoing demographic transition.

The policy comparison is between paths starting from the same inherited economy.

Thank You

Appendix

Parameter Restrictions

Parameter	Value	Lower	Upper	Position
β_{annual}	0.9900	0.94	0.99	Upper bound
κ_1	0.2794	0.02	50	Near lower bound
κ_C	0.3469	0.02	50	Near lower bound
χ	1.0480	0.1	5	Interior
H_0	8.1905	0.2	80	Interior
θ_0	0.0870	0	8	Interior
θ_1	0.0571	0.02	16	Near lower bound
F_1	0.1268	0	8	Interior
h_P	2.3000	0.1	2.3	Upper bound

Near-bound means within 1% of the allowed range; it does not establish a binding constraint. Fixed, normalized, and budget-derived objects are listed separately in the main table.

Steady-State Equilibrium

A steady state is a constant allocation, population structure, and price:

$$(N^*, H^*, G^*, \pi^*, P^*, r^*, T^*, \varpi^*).$$

Households optimize, persons and household distributions reproduce themselves, housing clears, and the fiscal rule holds.

- With no migration, births must replace deaths in a constant population.
- With migration, births and net inflows together offset deaths.
- A constant continuation value alone does not make a finite path stationary.

The Person-Cohort Law

Let $N_t^{g,a}$ denote resident persons of sex g and single year of age a .

The annual person law is

$$N_{t+1} = A_{t+1}N_t + f_{t+1}B_{t+1} + M_{t+1}.$$

- A_{t+1} ages existing cohorts using destination-age survival.
- $f_{t+1}B_{t+1}$ allocates newborn survivors by sex at age zero.
- M_{t+1} is the age–sex net-migration flow after survival.
- Model-period births are assigned to calendar cohorts before the annual law is iterated.

Births and household heads remain distinct objects throughout the transition.

The Household-Head Law

Fixed age–sex headship rates $\rho^{g,a}$ map persons into the target head stock:

$$H_{t+1}^{g,a} = \rho^{g,a} N_{t+1}^{g,a} = H_{t+1}^{\text{surv},g,a} + F_{t+1}^{g,a} - D_{t+1}^{g,a} + H_{t+1}^{M,g,a}.$$

The household operator supplies a raw within-age distribution $\tilde{G}_{t+1,a}$. Its mass is rescaled to the person-law head target:

$$G_{t+1,a}(x) = H_{t+1,a} \frac{\tilde{G}_{t+1,a}(x)}{\int \tilde{G}_{t+1,a}(x') dx'}.$$

- F and D are the minimum one-way formation or dissolution flow needed to attain $\rho^{g,a}$.
- Wealth, tenure, income, parity, and child-state composition are preserved within age.
- This is an accounting closure, not an estimated household-formation hazard.

Exact Annual Person Accounting

Let $s^{g,a}$ be survival and f^g the birth sex share. For newborns,

$$N_{t+1}^{g,0} = s_{t+1}^{g,0} f_{t+1}^g B_{t+1} + M_{t+1}^{g,0}.$$

For interior ages $a \geq 1$,

$$N_{t+1}^{g,a} = s_{t+1}^{g,a} N_t^{g,a-1} + M_{t+1}^{g,a}.$$

The terminal age is top coded, so survivors from the penultimate and terminal cells both enter the last cell. Every step satisfies

$$\sum N_{t+1} = \sum N_t + B_{t+1} - \text{newborn deaths} - \text{existing deaths} + \sum M_{t+1}.$$

The Within-Age Household Bridge

Let $\tilde{G}_{t+1,a}(x)$ be the raw household transition within age a , and let $H_{t+1,a}$ be the head mass implied by persons. Then

$$G_{t+1,a}(x) = \begin{cases} H_{t+1,a} \frac{\tilde{G}_{t+1,a}(x)}{\int \tilde{G}_{t+1,a}(x') dx'}, & \int \tilde{G}_{t+1,a} > 0, \\ H_{t+1,a} \Gamma_a(x), & \text{for an empty entrant age cell,} \end{cases}$$

where Γ_a is the externally specified entrant template. The operation exposes added and removed mass separately and moves no mass across ages.

A richer model would estimate economic states for newly formed, dissolved, and migrant heads rather than preserve the raw conditional composition.

Housing-Market Clearing

Let $q_t(x)$ denote the physical housing occupied after tenure choice. Market clearing requires

$$Q_t^H = \int q_t(x) dG_t(x) = H_0 \left(\frac{P_t}{\bar{P}} \right)^\eta.$$

For renters, $q_t = h_t^R$; for owners, $q_t = h_t'$. Choice probabilities are included when aggregating.

- The owner-service premium changes utility, not physical supply.
- Conditional renter policies enter demand only when renting is chosen.
- The same supply schedule applies to baseline and tax-reform paths.

Perfect-Foresight Solution

1. Guess the price and pension paths and compute rents from asset pricing.
2. Solve households backward using date-specific continuation values.
3. Advance household and population distributions from the inherited state.
4. Update prices and pensions until housing clears and the payroll budget balances at every date.
5. Extend the horizon and check that the measured outcomes are stable.

A stationary terminal value supplies a boundary condition; it does not establish stationarity of the simulated endpoint.